AI for Differentiated Learning

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Learning goal and core task

Levelly helps learners solve linear equations while choosing an appropriate amount of challenge and support. The shared core task is **3x + 6 = 21**, with solution **x = 5**. Changing the core equation updates all three paths together. The supported lesson domain uses positive whole-number solutions.

How differentiation works

Path	Adapted task	Guidance
Foundation	3x + 6 = 21	Direct inverse-operation hints; explicit subtract, divide, and check steps.
Standard	5x + 6 = 2x + 21	Variables on both sides; prompts encourage collecting terms and independent checking.
Advanced	Trail: C = 5h + 6; City: C = 2h + 21	Model a real comparison, find h = 5 and shared cost $31, then compare longer rentals.

The levels change representation, scaffolding, and the number of reasoning steps while preserving the underlying relationship. Immediate feedback distinguishes a wrong time from a correct time with an incorrect price.

AI model and prompts

A quantized open-source **SmolLM2-135M-Instruct** model runs through Transformers.js in a browser worker. No API key or server-side inference is required. The model receives the selected level, current task, three verified hints, and the learner's question. Its system prompt includes:

> Select the ONE hint that best matches the learner's question. A = getting started. B = making progress with the algebra. C = checking a result. Reply with exactly A, B, or C. Do not calculate or invent an answer.

The scaffolding instruction changes by level: direct concrete support for Foundation, independent algebra for Standard, and modelling and units for Advanced. The model selects a hint; the app displays the corresponding verified explanation. A second prompt chooses simpler, regular, or extended parameters from validated candidates; the selected core is transformed into all three paths. The AI can create a personalized practice set without inventing calculations. Invalid output is rejected.

Verification, privacy, and limits

The algebra engine determines all answers and worked steps. Tests check 2,000 generated examples across all levels, input boundaries, both advanced answers, and prompt/output validation. Desktop/mobile browser checks cover interaction, session progress, failure handling, and automated accessibility checks. The AI model may select an unhelpful hint; learners can reveal all verified hints. First use downloads model files, potentially over 100 MB. Questions stay local. In this cloud environment, model-host and publishing access are restricted; successful real-model inference and the public deployment need separate verification. No live result is claimed from failure-path tests.

Development disclosure: Code and report prepared with AI assistance for Maizhanov Zhalgas; review against course authorship rules before submission.